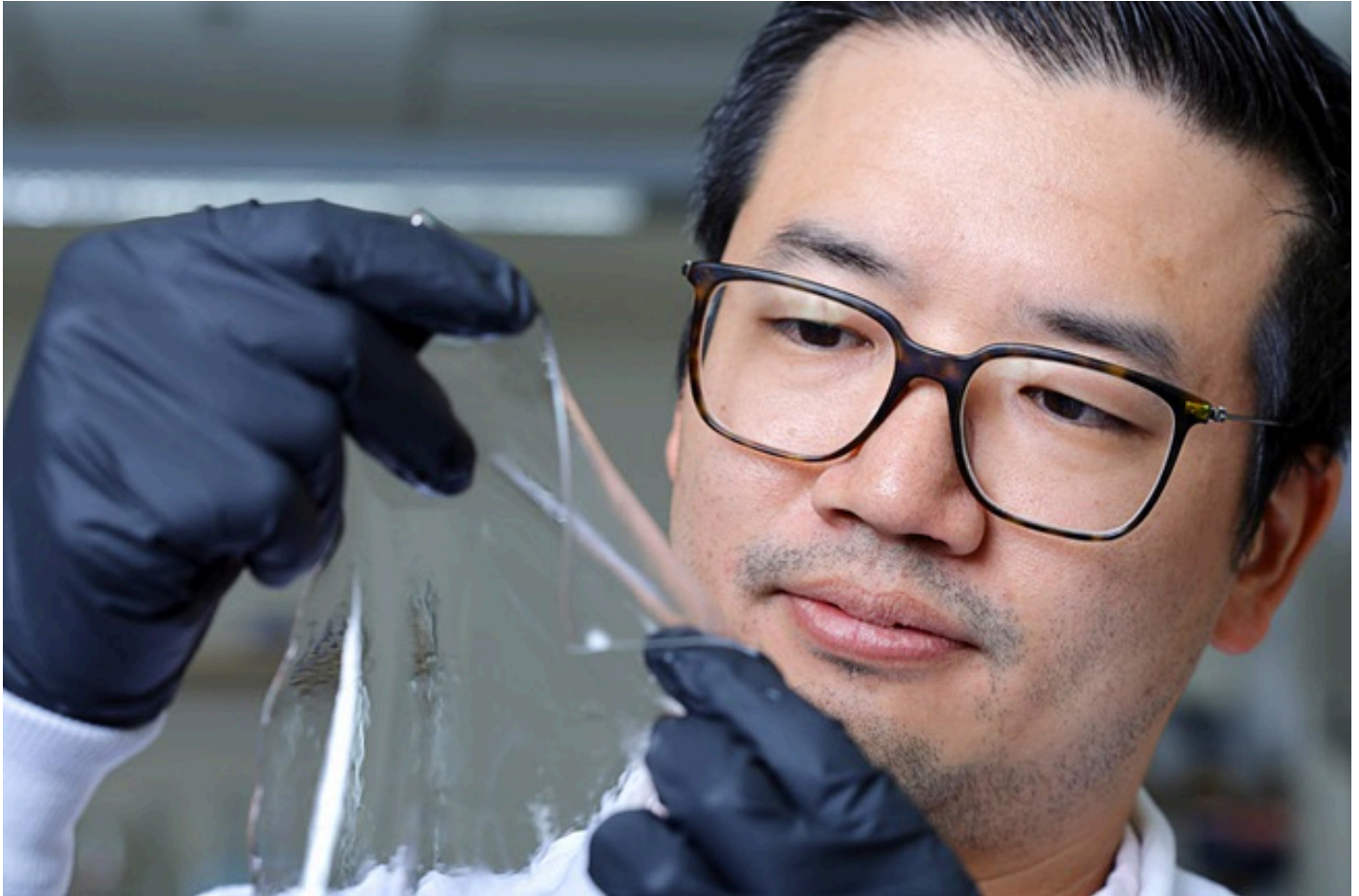




Harvesting water from air, UNLV engineer's device could have global appeal



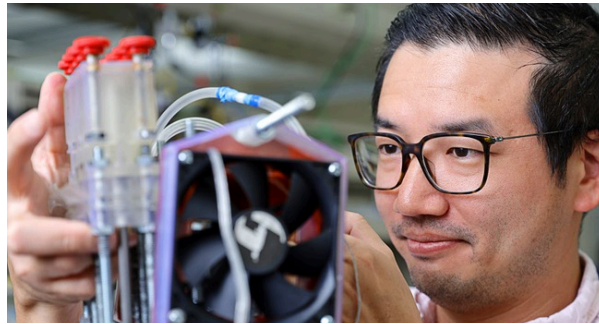
UNLV assistant professor Jeremy Cho holds a hydrogel "skin," used in his atmospheric water generator, in an UNLV Science and Engineering lab at UNLV Wednesday, Sept. 18, 2024. Photo by: [Steve Marcus](#)

By [Ilana Williams](#) ([contact](#))

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Drip, drip, drip.

That's the sound of Jeremy Cho's atmospheric water harvesting device extracting humidity from the air to make usable water in Da Kine Lab at UNLV.



VIEW GALLERY

UNLV Invention: Atmospheric Water Generator

The device, in its smaller state, looks like a box. One side has a small black fan and another has nuts and bolts with red caps sticking out of a translucent container.

There are a few screws and clear wires coming from the center.

It creates water by extracting humidity from the air. Cho's device can produce up to 17 liters per square meter per day in the lab.

"Just by being a civilized community, we're going to take some of that water," said Cho, an assistant professor at the UNLV department of mechanical engineering. "It's up to us to make sure that we use that water responsibly."

Cho started tinkering with the concept of atmospheric water harvesting as a science project, but soon he realized there could be technology for it, around the same time Lake Mead reached an all-time low in 2021, when inflow of water was at or below 11 million acre-feet per year.

"I took that as more reasons to really continue the work that I was doing," Cho said. "It became more than just a fun little science project."

He's hoping to mass produce the device commercially, where the ideal customer would be companies, organizations and agencies around the world that want to solve their water challenges and meet their water goals. Each device will fit the customer's needs — whether it be large enough for agriculture or small enough for an apartment complex.

Southern Nevada relies heavily on the Colorado River, with 90% of the city's water coming from it, and supplies 40 million people with water across the Lower Colorado Basin. Lake Mead levels have drastically declined from 95% full in 1999 to only 27% full in 2022. Nevada receives 300,000 acre-feet of water each year.

Water used indoors in Southern Nevada is recaptured, treated and returned to the Colorado River — and doesn't count toward the state's annual allotment.

However, water used for outdoor irrigation like watering grass or cooling air conditioner units can't be recycled.

That's where Cho's device comes in handy.

The system captures the condensate produced by an air conditioning unit and turns it into a sustainable solution through harvesting. Units, depending on the humidity, can produce five gallons overnight.

The device is an active research project, Cho stressed. The lab is constantly constructing and deconstructing the device for WAVR Technologies, the company he co-founded with Rich Sloan, the entrepreneur in residence at UNLV.

"Sustainability can also be applied to a community," Sloan said. "In Las Vegas, we have a vision to augment the very limited supply of water with a new and sustainable supply of water."

WAVR has different kinds of companies — a Fortune 500 beverage company, a medical company for in-home dialysis and a real estate company in Arizona — interested in Cho's device, Sloan said. They are also working with UNLV to become water neutral.

"I appreciate the many opportunities for this community that lie ahead as the community grows," Sloan said. "And I am respectful of the realities as we deal with water scarcity."

The concept of an atmospheric water harvesting device isn't new. Others are already in existence and readily available online.

Cho did not think about commercializing his device until it gathered up to 17 liters per square meter per day, or almost five gallons, in the lab and compared it with other atmospheric water harvesting companies, which can capture one or two liters per square meter per day, he said. Cho's device is also compatible with the low levels of humidity Southern Nevada experiences.

"We had to demonstrate that we could produce something larger," Cho said. "That motivated us to take our existing ideas of device design and to amplify the amount of (water)."

The current device the lab has been working on has been in development for six months. And that's just the beginning.

WAVR Technologies is raising funding from private investors, Sloan said, with plans to design, customize and manufacture atmospheric water harvesting devices for mass distribution.

Sloan envisions the technology being used in the Southwest where drought is prevalent, and states are jockeying for water rights from the Colorado River with Nevada.

It would also be valuable in humid parts of the nation like Florida and around the world in Northern Africa, the Middle East and islands that have water constraints.

“We hope to grow a really big business right here in Las Vegas,” Sloan said. “You’ve got the fun economy here with hospitality and sports and entertainment, but you also have a technology economy.”